

**(3D) Computer Vision Tasks:** Object Classification, Object Detection, Segmentation (Instance/Semantic/Panoptic), Depth Estimation, CNN-based Regression, Conditional GANs, Neural Rendering, Structure-from-Motion (SfM), 3D Scene Reconstruction, Mesh Generation, Point Cloud Registration, Simulation

## Segmentation / Depth Estimation

- Panoptic Segmentation: Fine-tuned **PanopticFCN** for pedestrian segmentation for ADAS/Autonomous Trains.
- Trained depth estimation models with consistency-aware losses and train/test-time augmentations to stress-test robustness.



## Neural Rendering

- Developed Neural Rendering pipelines using COLMAP for camera poses, performing point cloud registration, trained NeRFs/Gaussian Splats, and edited Gaussian Volumes in SuperSplats.
- Implemented a Tiny NeRF for forward rendering of road scenes, managing camera extrinsics.



## Robotic Simulation with NVIDIA Isaac Sim

- Generated synthetic data by simulating Robot in 3D Gaussian Volumes in Isaac Sim with MobilityGen and the Synthetic Data Generator, producing occupancy maps for robot navigation, and automated multi-sensor RGB-D and IR thermal camera data collection via the Sensor Processing Graph (SPG), streamed to ROS2 for real-time visualization.
- Converted fused 3D point clouds into meshes (via Marching Cubes, Poisson Surface Reconstruction, and Nvblox) as proxies for Gaussian Volumes.
- Implemented dynamic runtime 3D mesh generation in Unity3D.

## Fine-tuning Gaussian Volumes to Fill Blind Spots

- Fine-tuned 3D Gaussian representations by generating VLM scene captions and camera trajectories for novel view synthesis. Used video diffusion models to inpaint under-observed blind spots, distilling synthetic frames back into a second-stage Gaussian optimization to eliminate volumetric artifacts.

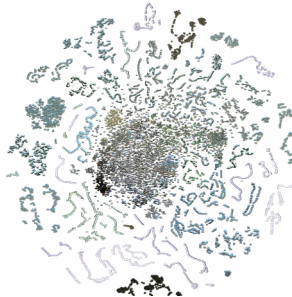
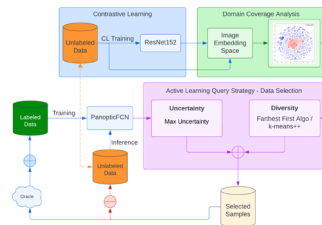


## Active Learning

- Performed Active Learning for Pan-optic Segmentation of pedestrians.
- Scaled test cases every cycle to maintain consistent train-test data distributions.
- Conducted evaluation on unseen test data in every cycle.

## Visualizing Image Embedding space for a sequence of images

- Extracted high-dimensional ResNet embeddings to visualize image sequences via t-SNE.
- Applied clustering (DBSCAN, OPTICS, Agglomerative) to automate Scenario Mining.
- Scenario Mining across urban cities, indoor/outdoor depots, countryside, forests, and open fields, identifying edge cases, such as pedestrians in tunnels and diverse pedestrian activities under different lighting.



## Contrastive Learning

- Trained a Siamese ResNet-152 backbone using NT-Xent loss to learn robust image representations.
- Mapped feature space to cluster semantically similar images while distancing dissimilar samples.
- Used learned embeddings for downstream tasks, including diverse data sampling.

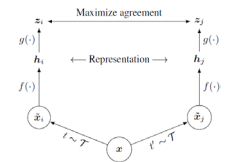
## Uncertainty Estimation

- Standard Strategies: Margin Sampling, Least Confidence, Entropy Criterion.
- Explored Methods: Query-by-Committee (QBC), Monte-Carlo (MC) Dropout, Test-Time Augmentation (TTA), Ensemble Methods.
- QBC with TTA workflow utilizing KL-Divergence to quantify distances between predictive probability distributions.

Fine-tuned Representation Space

## Diversity Query Strategy

- Fine-tuned a high-dimensional Embedding space with **Contrastive Learning**
- High-dimensional features are mapped to 2D t-SNE space.
- Diversity Sampling: Used Farthest First/Gonzalez Algorithm for querying samples.



## (conditional) GANs

- Developed GANs and conditional GANs (cGANs) from scratch
- Utilized CelebA dataset to generate images with specific attribute combinations.



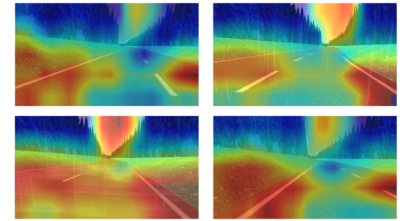
## CNN-based Regression for Autonomous Driving

- Self-Driving Car:** Trained a 5-layer CNN architecture to predict steering angles and perform road-line detection.
- Control Logic: Used predicted steering angles to determine vehicle velocity.



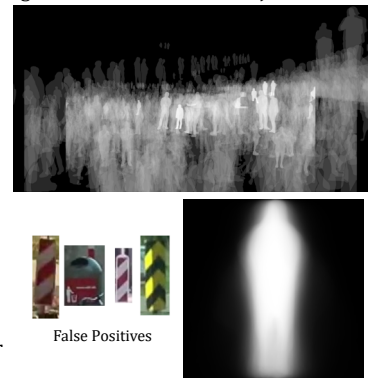
## Grad-CAM Analysis

- Performed Grad-CAM analysis to identify pixels driving steering angle predictions.
- Analyzed model behavior in Out-of-Domain scenarios such as rain, fog, and night sky.
- Evaluated model reactions under domain shift.



## Image/3D Point Cloud Processing

- Analyzed objects based on location and shape utilizing heatmaps.
- Calculated heatmaps by taking the logarithm of the detected objects.
- Filtered False Positives from inference results before passing to domain experts for labeling.
- Used pixel-wise distributions and Euclidean distance to extract top-3 matching and non-matching samples per image for Contrastive Loss.
- Verified 3D point cloud accuracy by performing Euclidean distance benchmarks.
- Managed 3D data pipelines by converting COLMAP binary (.bin) outputs to .ply and text formats for cross-tool processing.

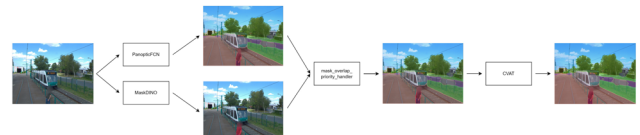


## Data Curation

- Semantic Retrieval: Performed data curation for specific scenario retrieval across massive unlabeled datasets.
- Used Zero-shot Transfer for all data curation and retrieval tasks.
- Vision Transformers: Used BLIP/BLIP-2 for image-text semantic retrieval.

## Semi-Automatic Annotation and Data Labeling Policy

- Automate the Annotation pipeline for faster processing and training.
- Integrated SAM Docker within CVAT for native AI-assisted labeling.
- Used YOLO, SAM, MaskDINO, Mask2Former, and PanopticFCN for semi-automatic annotation.



- Defined class taxonomy and labeling guidelines, addressed edge cases for consistency.
- Handled RLE to Polygon format conversion and vice-versa.
- Conducted annotation sessions using CVAT.

## Evaluation Metrics

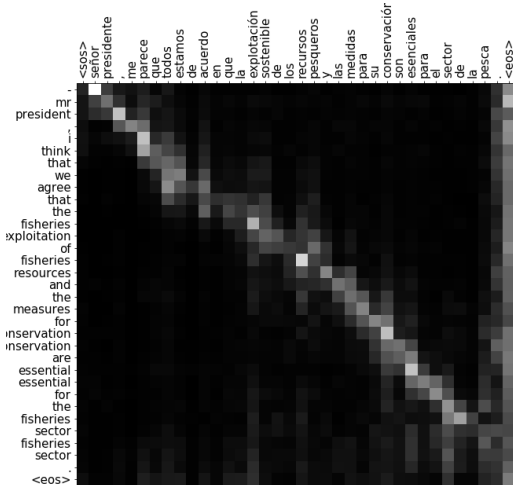
- Average Precision (COCO) and Pedestrian Detection Rate
- Inception score (IS) and Fréchet Inception Distance
- PSNR
- Qualitative Analysis
- In-Domain vs Out-of-Domain image evaluation
- Spatial Analysis: Bounding box height vs Width ratio, Segmentation mask area count.



**Natural Language Processing Tasks:** Neural Machine Translation (NMT), Text-to-Code Generation, AI Agent, Fine-Tuning LLMs, Retrieval Augmented Generation (RAG), Sentiment Analysis, Text Extraction (OCR)

### Neural Machine Translation

- Spanish → English Translation
- Developed an Encoder-Decoder architecture with Attention mechanism from scratch
- Trained and evaluated on parallel corpora, achieving a BLEU score of 25.37



### Hierarchical Attention Network

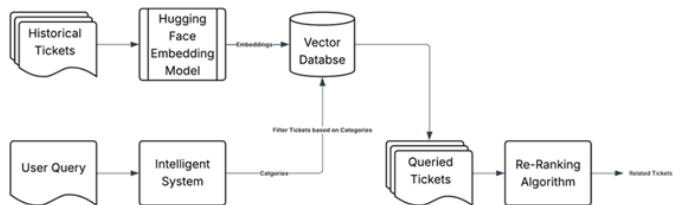
- Document Classification
- Implemented a custom Encoder architecture with hierarchical attention mechanisms
- Designed separate word-level and sentence-level attention layers from scratch

sentence\_1 74milipeachbiden  
 sentence\_2 **can't** how people do **not** want police until they need them  
 sentence\_3 i **remember** them playing with the **hair** on my legs  
 sentence\_4 back pedal file joe  
 sentence\_5 joe  
 sentence\_6 awful funny **news** will dub the climate as in crisis but will not address the boarder crisis as such

- Visualized attention weights to interpret model decisions (darker red = higher word importance, darker blue = higher sentence importance)

### Ticket Checker

- Developed a deep learning-based multi-class ticket classification model
- Applied category-based filtering on a vector database using predicted labels
- Implemented a Retrieval-Augmented Generation (RAG) pipeline
- Generated ticket embeddings using HuggingFace SentenceTransformers
- Retrieved semantically similar tickets from a vector database
- Apply re-ranking algorithm on the retrived tickets
- Applied a re-ranking algorithm to improve retrieval relevance and response quality



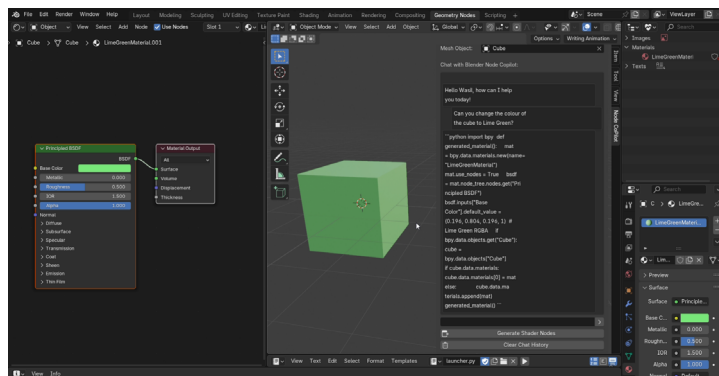
### Theoretical Research on Search and Recommendation Systems

- Self-directed study of two-stage industry retrieval and ranking pipelines.
- Offline Retrieval:** Training Two-Tower Bi-Encoders via Contrastive Learning and Hard Negative Mining (clicked vs. non-clicked items).
- Indexing item embeddings in a Vector DB using ANN for scale.
- Pre-computing item embeddings and indexing them in a Vector Database using ANN (Approximate Nearest Neighbors) algorithms.
- Online Retrieval:** Embedding user queries into high-dimensional vectors through the trained Query Encoder.
- Executing a low-latency ANN search against the indexed item store to retrieve the Top-K most relevant candidates, effectively narrowing the search space for the high-precision ranking stage.
- Offline Ranking:** Architecting a Feature Store to manage User and Items.
- Training Neural Networks for CTR (Click-Through Rate) Prediction and intent detection via Pointwise/Pairwise loss.

- Online Ranking:** Dynamically augmenting the Top-K candidates with real-time features and historical state fetched from the Feature Store.
- Scoring enriched candidates via the Ranking Model and applying final sorting.

### Blender Text-to-Code

- Developed a Text-to-Code ChatBot within Blender UI
- Translated natural language prompts into Geometry Nodes and Shader Node configurations
- Built an in-application chatbot interface inside Blender to enable real-time communication between the LLM and the 3D environment
- Integrated AWS-hosted foundation models for inference
- Fine-tuned a LLaMA-based model using Reinforcement Learning from Human Feedback (RLHF) with Proximal Policy Optimization (PPO)

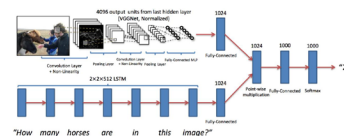


### Quality Agent

- AI-Powered Requirement Analysis System
- Developed a requirement analysis and validation tool using a multi-agent AI framework
- Designed specialized agents for: Automated test case generation, Requirement analysis and validation, Data extraction from Microsoft Word documents, Repository data retrieval via SVN API
- Integrated a self-service portal using JavaScript for user interaction and workflow automation

### Vision + Language

- Built a multimodal VQA system integrating visual features with textual embeddings
- GloVe word embeddings for textual feature representation
- Compared LSTM, GRU, and Bidirectional architectures for sequence modeling



### Table Extraction

- Extracted tables from PDFs using Amazon Textract and Azure Document Intelligence
- Converted table structures into Excel format for analysis
- Transformed tabular data into textual representations

### Sentiment Analysis Projects

- Kaggle Competition: Quora Insincere Questions Classification and Bags of Words Meets Bags Of Popcorn.
- Utilized GloVe and Google Word2Vec embeddings for feature representation

### Evaluation Metrics

- Accuracy
- BLEU Score
- F1 Score
- Area under the ROC curve (AUC)
- Qualitative Analysis